

Class

10

Roll No.

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Mega Vedantu Scholarship Admission Test(MVSAT)

Question Booklet

Time Allowed

1 Hour, 30 Minutes

Question Booklet Set

Max. Marks

180

A

Mental Ability, Maths and General Science

Welcome, Superstar! 🌟

Before you dive into cracking this test, let's go over some super important guidelines to help you get through this tension-free!

Ready? Here we go!

- 1. Help Us Identify You** – Before you dive into solving our fun set of questions, **write down your Roll Number in the box at the top right of the question booklet and also on the OMR.** Read the **Important Instructions** on the OMR and make sure you fill in your details—**Name, School Name & City, Parent Name, Parent Mobile Number, Current Grade, Question Paper Set, Target, Board, and Signature.** These details ensure that your answer sheet is counted. If you miss something, your answers might vanish into thin air! So, double-check like a pro.
 - 2. Best of Four** – Every question has four options, but only one is the right answer. Pick wisely, and mark your choice with your **Black Ball Pen.** Be confident, be bold!
 - 3. Cross-check the Question Paper** – Your Question Booklet has 45 questions (plus a page of instructions like this). If it feels incomplete or weird, **call your room invigilator ASAP in the first 10 minutes.** Time's ticking, no pressure!
 - 4. Subject Breakdown** –
 - **Mental Ability:** Q. Nos. 1 – 20
 - **Maths:** Q. Nos. 21 – 30
 - **General Science:** Q. Nos. 31 – 45
 - 5. Make Sure Your Answer Counts** – Each correct answer earns you 4 marks. There is no negative marking for incorrect answers. So, relax and go all in, don't leave anything blank!
 - 6. Stay Gadget-free** – As cool as they are, **gadgets are banned in the exam hall.** No calculators, phones, or tech toys. Today, it's all about using your brainpower!
 - 7. The Paper Should Be in One Piece** – Your Question Booklet is a complete set, so don't tear out any pages. When the clock runs out, hand in your OMR sheet to the room invigilator.
 - 8. Leave When Time Runs Out** – Once you're in, there's **no going out until the exam ends.** Stay cool, focus, and give it your best till the last second!
- Now that you're all set, take a deep breath, give this paper your best shot, and remember:

You've got this! 💪🌟

$$OB = \sqrt{SB^2 - SO^2} = \sqrt{8^2 - 4^2} = \sqrt{64 - 16} = \sqrt{48} = 4\sqrt{3}$$

$$S_{\triangle ABC} = \frac{1}{2} SO \cdot AB = SO \cdot OB = 4 \cdot 4\sqrt{3} = 16\sqrt{3}$$

Mental Ability

Question 1: An office staff works for 4 days consecutively, then has the next day off; he works for 4 more days and has a day off on the next day; and so on. Today is his day-off and it is a Sunday. The minimum number of days the staff must work to have his off-day as Sunday is _____



Options:

(a) 28

(b) 20

(c) 22

(d) 24

Question 2: Given below is the triangular form of AMTI.

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      A
     A M A
    A M T M A
   A M T I T M A
  
```

The number of ways you can spell AMTI, top to bottom, right to left or left to right or a combination of these is _____.

Options:

(a) 12

(b) 13

(c) 15

(d) 16

Rough Work

Question 3: 🧠 The Cryptic Age Conundrum at the Addams Mansion 🦇

One stormy evening, little Pugsley Addams, age 8, was tinkering with his guillotine toy when he turned to Morticia, his elegant and mysterious mother, and asked:

"Mother, how old are you really? Not in vampire years—your actual age!"

Morticia, sipping her nightshade tea, replied with a cryptic smile:

"Darling, when you are as old as I am now, I shall be 54 years old."

Pugsley blinked. The bats fluttered. Even Things paused mid-creep.

🌿 Your Challenge:

Can you deduce Morticia Addams's current age, based on her eerie riddle?

Options:

- (a) 31 (b) 32 (c) 33 (d) 34

Question 4: Starting from the number 1, Ritu generates a series of numbers as 1, 3, 6, 11, 18, 29, 42, ... such that the differences of the consecutive numbers from the beginning give consecutive primes. In this series she came across a perfect square for the first time. The Square root of this perfect square is

Options:

- (a) 14 (b) 19 (c) 23 (d) 21

Question 5: Pointing to a photograph. Bajpai said, "He is the son of the only daughter of the father of my brother." How is the man in the photograph related to Bajpai?

Options:

- (a) Maternal uncle (b) Nephew (c) Son (d) Brother

Question 6: Six children are playing Hide & Seek in a park and were standing at different positions such that P is 9 m towards the north of T, Q is 4 m to the west of O, who is 12m to the east of T. U is 6 m to the south of Q, R is 3 m to the north of O. Then few of them moves and their new positions are denoted by adding (i) to their respective names. P moved 5m towards east. O moves 3 m towards the east. Q moves 5 m towards the north.

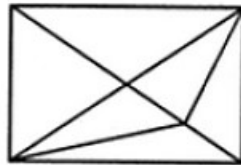
What is the shortest distance between T and U?

Options:

- (a) 9m (b) 11m (c) 10m (d) 14m

Rough Work

Question 7: Find the number of triangles in the given figure.

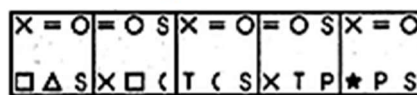


Options:

- (a) 11 (b) 13 (c) 15 (d) 17

Question 8: Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

Problem Figures:



- (A) (B) (C) (D) (E)

Answer Figures:



- (1) (2) (3) (4)

Options:

- (a) 1 (b) 2 (c) 3 (d) 4

Question 9: Six persons visit a place on two dates 11 and 22 in three different months i.e., September, October and November. They all like different colors viz. white, blue, pink, green, red and yellow but not necessarily in the same order as given. The one who likes yellow visits third position after L.

L and S visit on same date and in the month of even numbered days. Two persons visit between S and R who likes green. Number of persons visit before R is same as the number of persons visit after V. The one who likes white visits just before H. No one visit between K and the one who likes blue. The one who likes red visits before the one who likes pink.

Which of the following colours does H like?

Options:

- (a) Red (b) Blue (c) yellow (d) white

Rough Work

Question 10: 🕒 Hermione's Enchanted Path Through the Forbidden Forest 🌲



One crisp evening, Hermione Granger stood at Point A, just outside the edge of the Forbidden Forest, facing north. She was on a mission to retrieve a rare potion ingredient hidden deep within the woods.

Here's how her enchanted journey unfolded:

1. She walked 5 km north, wand at the ready, dodging mischievous bowtruckles.
2. Then she turned 90° to her right (now facing east) and walked another 5 km, arriving at Point B, where she spotted a glowing unicorn hoofprint.
3. She turned 90° again (now facing south) and walked 3 km, brushing past whispering vines.
4. One final 90° turn (now facing west) and walked 3 km, reaching Point C, where she found the shimmering Moonshade root.

🧠 Your Challenge:

From her original starting point A, what is the direction of Point B and Point C, respectively?

Options:

- (a) East, east (b) East, North-East (c) North-East, East (d) North- East, North-East

Rough Work

Question 11: A coin is in a fixed position. Another identical coin is rolled around the edge of the first one. How many complete revolutions will be made by the revolving coin before it reaches its starting position?



Options:

- (a) 1 (b) 2 (c) 3 (d) 4

Question 12: Pointing to a photograph, a woman said, "This man's Son's sister is my mother-in-law". How is the woman's husband related to the man in the photograph?

Options:

- (a) Son (b) Nephew (c) Grandson (d) Son-in-law

Question 13: What will be next term in the series 3, 8, 13, 24, 41,

Options:

- (a) 65 (b) 75 (c) 70 (d) 80

Question 14: If MENTAL : SMXFOB then ABILITY: _____

Options:

- (a) GJSXWJQ (b) GSXWJJQ (c) SGXWJJQ (d) SJXQJWG

Question 15: In a coded language.

'Busy bees' are coded as 'Cpu cff'

'Busy crows' are coded as 'Cpu hup'

'Bright crows' are coded as 'Csj hup'

Then busy crows are clever will be coded as _____

Options:

- (a) Cpu Hup bsf dmf (b) Cpu hup bsf Dmf (c) cpu Hup Bsf Dmf (d) cpu hup bsf Dmf

Rough Work

Question 16: Five friends A, B, C, D and E are standing in a row facing south but not necessarily in the same order. Only B is between A and E, C is immediate right to E and, D immediate left to A. On the basis of above information, which of the following statements is definitely true?

Options:

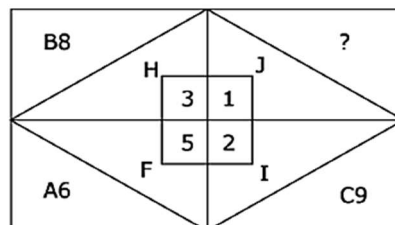
- (a) B is to the left of A (b) B is to right of E
(c) A is second to the left of E (d) D is third to the left of E

Question 17: If MENTAL = 390, ABILITY = 546, REASONING = 918, then COMPETENCY = ?

Options:

- (a) 782 (b) 842 (c) 1190 (d) 1340

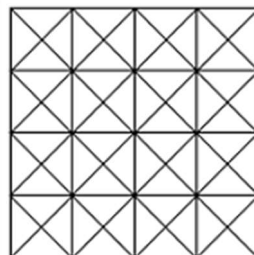
Question 18: Which expression will replace the question mark?



Options:

- (a) E 7 (b) D 10 (c) D 7 (d) E 10

Question 19: What is the total number of squares in the given figure?



Options:

- (a) 62 (b) 72 (c) 82 (d) 92

Question 20: If + stands for $\times$, - for $\div$, $\times$ for - and $\div$ for +, find the value of $26 + 74 - 4 \times 5 \div 2$

Options:

- (a) 220 (b) 376 (c) 478 (d) 488

Rough Work

Mathematics

Question 21: If

$$S_1 = \{1\}$$

$$S_2 = \{2, 3\}$$

$$S_3 = \{4, 5, 6\}$$

$$S_4 = \{7, 8, 9, 10\}$$
 -----so on find out the sum of all the elements of S_{23} ?

Options:

(a) 6093

(b) 6095

(c) 5092

(d) 4295

Question 22: 🦾 **Avengers Scenario-Based Math Problem:**



From the top of the Avengers Tower, **Tony Stark** is monitoring incoming threats using his suit's advanced targeting system. He spots a hostile alien scout speeding straight toward the tower on a hover-bike at a **uniform velocity**.

Initially, his **HUD (Heads-Up Display)** shows the **angle of depression** to the scout is **30°**. After **12 minutes**, the angle of depression increases to **45°**.

Assuming the alien continues at the same speed and in a straight line toward the base of the tower...

🔍 **How much longer will it take before the alien reaches the base of Avengers Tower, triggering the tower's automatic defense system?**

Options:

(a) 16 min 23 seconds

(b) 23 min 16 seconds

(c) 16 min 39 seconds

(d) 39 min 16 seconds

Rough Work

Question 23: If the co-ordinates of the mid-points of the sides of a triangle are (1, 1), (2, -3) and (3, 4), then centroid of triangle is :

Options:

(a) $\left(3, \frac{1}{3}\right)$

(b) $\left(1, \frac{2}{3}\right)$

(c) (3, 1)

(d) $\left(2, \frac{2}{3}\right)$

Question 24: If $\operatorname{cosec} 39^\circ = x$, the value of $\frac{1}{\operatorname{cosec}^2 51^\circ} + \sin^2 39^\circ + \tan^2 51^\circ - \frac{1}{\sin^2 51^\circ \sec^2 39^\circ}$

Options:

(a) $\sqrt{x^2 - 1}$

(b) $\sqrt{1 - x^2}$

(c) $x^2 - 1$

(d) $1 - x^2$

Question 25: The value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 179^\circ$ is

Options:

(a) $\frac{1}{\sqrt{2}}$

(b) 0

(c) 1

(d) -1

Question 26:   **Mission: Galactic Fireworks**

Scenario:

To celebrate the anniversary of Earth's alliance with the Nova Corps, **Stark Industries** launches a prototype rocket from a secret launch pad in Wakanda.

The height H (in meters) of the rocket above the ground after 't' seconds is modeled by the equation:

$$H(t) = -5t^2 + 20t + 10$$

Tony Stark is monitoring the launch from orbit and needs to know:

 **At what time will the rocket reach its maximum height, and what is that height?**

If the rocket doesn't reach the calculated peak, the midair fireworks will fail — and Thor is **not** going to be happy about missing the show.

Options:

(a) 2, 30

(b) 5, 60

(c) 1, 15

(d) 3, 90

Question 27: If $217x + 131y = 913$ and $131x + 217y = 827$, then the value of $x + y$ is :

Options:

(a) 8

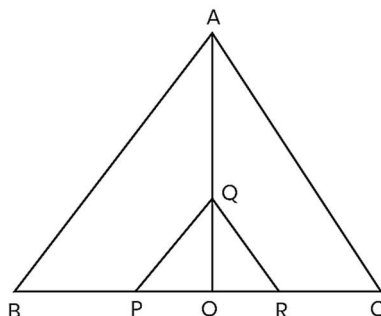
(b) 5

(c) 7

(d) 6

Rough Work

Question 28: In the given figure, $PQ \parallel BA$ and $RQ \parallel CA$. If $BP = 7$ cm, $RC = 21$ cm and $PO = 4$ cm, then RO is



Options:

- (a) 12 cm (b) 15 cm (c) $\frac{4}{3}$ cm (d) None of these

Question 29: 🎲 🧙 **Mission: Dice of Destiny**



Scenario:

In the enchanted vault beneath **Doctor Strange's Sanctum Sanctorum**, a mystical puzzle protects a hidden relic. The puzzle involves two magical dice that glows when thrown together.

To unlock the relic's seal, both dice must show the **same number** when rolled together.

What is the probability of getting the same number on both dice to unlock the relic's seal ?

Options:

- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{1}{12}$

Rough Work

Question 30: HCF and LCM of expressions $(x^3 - 1)$ and A are $(x - 1)$ and $(x^6 - 1)$ respectively, then value of A is:

Options:

- (a) $x^3 + 1$ (b) $x^4 - x^3 + x - 1$ (c) $(x - 1)(x^2 - x + 1)$ (d) $(x - 1)(x^2 + x + 1)$

General Science

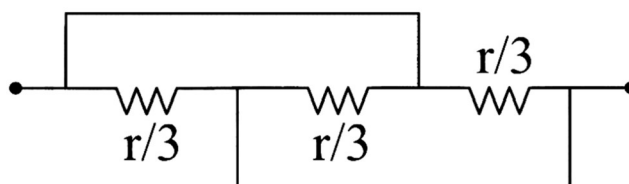
Question 31: Imagine you're building a futuristic city powered entirely by clean energy. To kickstart the entire power grid, **you need a device that can generate electric current from scratch.**

Which of the following should you install at the heart of your energy station?

Options:

- (a) Generator (b) Galvanometer (c) Ammeter (d) Motor

Question 32: Find the equivalent resistance between two ends of the following circuit



Options:

- (a) r (b) $\frac{r}{6}$ (c) $\frac{r}{3}$ (d) $\frac{r}{9}$

Rough Work

Question 33: While training his Rasengan, Naruto uses a special mirror made by Tenten. He stands 10 cm away from it, and needs to see a magnified image of his face at a comfortable viewing distance of 25 cm.

To make this chakra mirror work perfectly, **what should be its radius of curvature?**

(Hint from Shikamaru: Use the mirror formula)



Options:

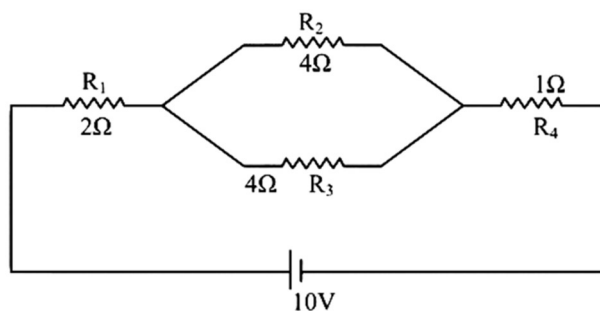
- (a) 60 cm (b) 24 cm (c) -60 cm (d) -24 cm

Question 34: Light travels in two media M_1 and M_2 with speeds $1.5 \times 10^8 \text{ ms}^{-1}$ and $2.0 \times 10^8 \text{ ms}^{-1}$ respectively. The critical angle between them is:

Options:

- (a) $\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$ (b) $\tan^{-1}\left(\frac{2}{3}\right)$ (c) $\cos^{-1}\left(\frac{3}{4}\right)$ (d) $\sin^{-1}\left(\frac{2}{3}\right)$

Question 35: In the given circuit, the current in resistance R_3 is:



Options:

- (a) 2 A (b) 1.5 A (c) 1 A (d) 2.5 A

Rough Work

Question 36: In Snape's Potions class, powdered sodium chloride and concentrated sulphuric acid lay ready on the desk. With a flick of his wand, Snape let the acid drip onto the salt. At once, dense white fumes rose with a sharp, choking smell.

Harry whispered, "That pungent odour—it must be hydrogen chloride gas."

Hermione nodded knowingly.

Ron, coughing, pointed to another tube filled with white powder. "What's that for?"

Snape turned to face him. "Ah, Mr. Weasley, you've noticed the guard tube. Can anyone tell me what it contains—and why?"



Options:

- (a) Compound – Water (H_2O); Purpose – To dissolve HCl gas before collection
- (b) Compound – Sodium hydroxide (NaOH); Purpose – To neutralize any escaping HCl gas
- (c) Compound – Potassium hydroxide (KOH); Purpose – To cool down the gas before collection
- (d) Compound – Calcium chloride (CaCl_2); Purpose – To absorb atmospheric moisture and prevent it from entering the setup

Question 37: Generally, metallic oxides are basic, but some are amphoteric in nature. Out of the given metallic oxides, how many are amphoteric—

[ZnO , MgO , Ga_2O_3 , PbO , CaO , MnO , FeO]

Options:

- (a) 2
- (b) 3
- (c) 4
- (d) 5

Rough Work

Question 38: In diamond, an allotrope of carbon, carbon atoms are arranged-



Options:

- (a) In hexagonal rings (b) Trigonal fashion (c) Tetrahedral fashion (d) Saucer ball like

Question 39: Which of the following is a redox reaction?

Options:

- (a) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (b) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
(c) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$ (d) $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$

Question 40: When a metal reacts with sulphuric acid, the products formed are hydrogen gas and metal sulphate. What do you think of the products formed, if nitric acid is used in place of sulphuric acid?

Options:

- (a) Metal sulphate, NO_2 (b) Metal nitrate, SO_2 (c) Metal sulphate, SO_2 (d) Metal nitrate, NO_2

Question 41: Why does a person suffering from chronic kidney failure experience increased blood pressure?

Options:

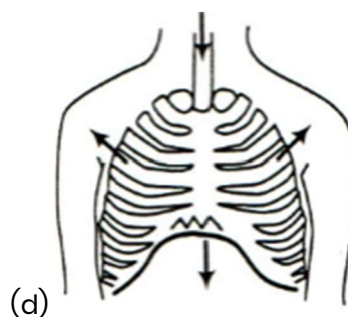
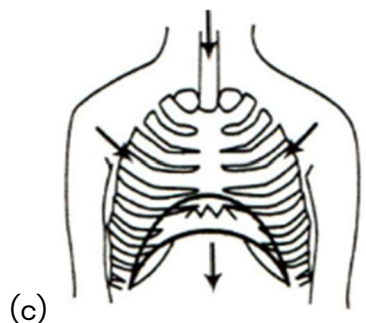
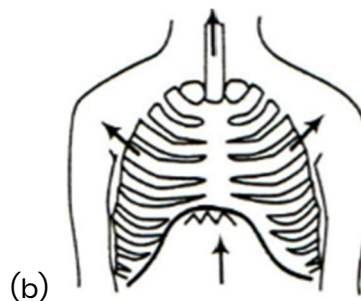
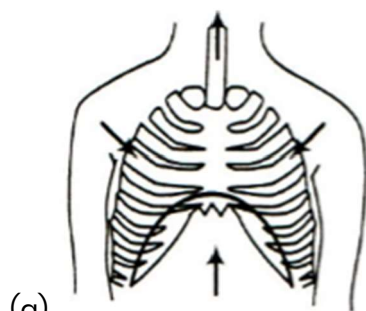
- (a) The kidneys produce excess renin, increasing blood pressure
(b) The blood vessels constrict due to reduced kidney function
(c) The kidneys retain excessive water and sodium, increasing blood volume
(d) Blood flow to the brain increases, causing pressure to rise

Rough Work

Question 42: Exhalation is the process of expulsion of air through respiratory tract.

Which of the following figure does illustrate the process of exhalation?

Options:



Rough Work

Question 43: The “Body Village” is under attack by the Evil Toxin Clan!

Naruto, Luffy, and Deku rush to the Kidney Fortress for the first line of defense.

Naruto yells: “We’ve gotta clean the blood fast! Which guardian carries it to the main gate ...A...?”

Luffy stretches and says: “And what do we call this epic cleansing move? Oh yeah—...B...!”

Deku, pulling out his hero notebook, adds: “According to All Might’s biology class, about ...C... mL of blood is filtered here every minute, which equals ...D... of the heart’s epic output!”



Options:

- (a) A–Glomerulus; B–filtration; C–800–900; D–1/4th
- (b) A–Glomerulus; B–filtration; C–1100–1200; D–1/5th
- (c) A–Glomerulus; B–Reabsorption; C–1100–1300; D–1/6th
- (d) A–Loop of henle; B–filtration; C–1100–1500; D–1/5th

Question 44: If a drug inhibits the reuptake of a neurotransmitter at the synapse, what will likely happen to the signal transmission?

Options:

- (a) Signal transmission will weaken because the neurotransmitter will degrade.
- (b) Signal transmission will stop entirely due to lack of neurotransmitter recycling.
- (c) Signal transmission will strengthen because the neurotransmitter remains in the synapse longer.
- (d) Signal transmission will become erratic as neurotransmitter production increases.

Question 45: In the Modi Mansion garden, Gopi Bahu excitedly watered her rose plant, which suddenly started growing taller than Ahem’s ego. Kokila sternly asked,

“Gopi, which plant hormone is making these roses grow as if they’re running a marathon?”

Options:

- (a) Auxin
- (b) Gibberellin
- (c) Cytokinin
- (d) Ethylene

Rough Work